

UNIT – 1
ASSESSMENT – 1
PROBLEMS

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1. Aim:

To solve the Water Jug Problem using Artificial Intelligence problem-solving techniques.

Problem Statement:

You are given two jugs:

- One **4-gallon jug**
- One **3-gallon jug**

Neither jug has any measuring marks. There is a water pump that can fill either jug completely.

The following operations are allowed:

- Fill a jug completely.
- Empty a jug completely.
- Pour water from one jug into the other until either:
 - the source jug becomes empty, or
 - the destination jug becomes full.

Determine the sequence of actions required to obtain **exactly 2 gallons of water in the 4-gallon jug.**

Assumptions

1. Water is available in unlimited quantity.
2. Water may be poured onto the ground.
3. No measuring device other than the two jugs is available.
4. Water can be transferred freely between the jugs.

2. Aim:

To study how a Mars Rover functions as an intelligent agent and analyze its percepts, operating environment, actions, performance measures, and suitable agent architecture.

Problem Statement:

A Mars Rover is an autonomous robotic vehicle developed to explore the surface of Mars. It performs scientific experiments, collects soil and rock samples, captures images, and sends data back to Earth.

3. Aim:

To explore the search space using search strategies and formulate the problem components for the **8-Queens Problem**.

Problem Statement:

Place **8 queens** on an **8 × 8 chessboard** such that **no two queens attack each other**.

A queen can attack another queen if they are:

- In the same row
- In the same column
- In the same diagonal

The objective is to find a valid arrangement where all 8 queens are placed safely.

4. **Aim:**

To identify the type of problem-solving agent used in the OLA Cab Booking application and write the pseudocode for the cab selection process.

Problem Statement:

A customer wants to travel from one location to another using the **OLA Cab Booking** mobile application. The customer can choose from different cab types such as **Mini, Micro, Sedan, Shared, Prime**, etc., depending on comfort, availability, and cost.

Identify the type of **problem-solving agent** that can be used and write the **pseudocode** for the above problem.

5. Aim:

To find the least-cost path from the starting warehouse **S** to the destination warehouse **G** using the **Uniform Cost Search (UCS)** algorithm.

Problem Statement:

A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse **G** using the **Uniform Cost Search (UCS)** algorithm.